Dealt with solving linear congruences.

$$ax \equiv b \pmod{n}. \quad (*)$$

$d = \gcd(a,n)$ and $d \mid b$ then $(*)$ has d solutions as described in theorem 6.3.

CRT. For pairwise coprime moduli $m_1, \dots, m_k$. CRT says that a set of congruences.

$$x \equiv b_i \pmod{m_i} \quad i = 1, \dots, k.$$

will correspond to a unique congruence dens modulo $M = m_1 m_2 \cdots m_k$.

Now focus on how to find solutions to the general problem

$$f(x) \equiv 0 \pmod{n}$$

Use prime factorization of $n = \prod_{i=1}^{r} p_i^{a_i}$

Consider the set of congruences.

$$f(x) \equiv 0 \pmod{p_i^{a_i}}, \quad i = 1, \dots, r.$$

In theorem 7.2 we'll see to how to solve each of these.

Then when we take any combination of solutions.

$$(*) \quad x \equiv b_i \pmod{p_i^{a_i}} \quad i = 1, \dots, r$$

ie $f(b_i) \equiv 0 \pmod{p_i^{a_i}}$

Can use the C.R.T. to find the unique solution x modulo $n = \prod_{i=1}^{r} p_i^{a_i}$ corresponding to the set of solutions in $(*)$

We would do this for each different combination of solutions ~~to~~ in $(*)$

First result: Lagrange's Theorem.

<u>Theorem 7.1</u> Let f be a poly with integer coefficients $f(x) = \sum_{i=1}^{n} a_i x^i$ $(a_i \in \mathbb{Z})$

If $\deg f$ is n, ie. $a_n \not\equiv 0 \mod p$, ie. $p \nmid a_n$

then the poly. cong.

$$f(x) \equiv 0 \pmod{p}$$

has <u>at most</u> n solutions modulo p.

<u>Proof</u> (Nice substantial argument, using proof by induction and proof by contradiction).

We perform induction on n, the degree of f.

Base case <u>$n=1$, ie. linear poly. congs.</u>

$$a_1 x + a_0 \equiv 0 \pmod{p}$$

Since $p \nmid a_1$, $\gcd(p, a_1) = 1$, ~~so~~ there's

a single solution

$$x \equiv -a_1^{-1} a_0 \pmod{p}.$$

So the theorem is true for $n=1$.

So now we assume theorem is true for all polys of degree k, for some $k \geq 1$.

So now we must prove the theorem for all polys of degree $k+1$. This we will do by using proof by contradiction.

So we assume is false for $k+1$, i.e. assume there exists at least one "bad" poly g of degree $k+1$

i.e. $g(x) = \sum_{i=1}^{k+1} a_i x^i$, $p \nmid a_{k+1}$

and $g(x) \equiv 0 \pmod{p}$ has at least $k+2$ solutions

$$x \equiv x_0, x_1, x_2, \ldots, x_{k+1}$$

We will construct from g a 'bad' poly h of deg k, i.e. one having at least $k+1$ solutions

<u>Construction:</u> Consider the new poly G.

$$G(x) = g(x) - g(x_0)$$

$$= \sum_{i=1}^{k+1} a_i x^i - \sum_{i=1}^{k+1} a_i x_0^i$$

$$= \sum_{i=1}^{k+1} a_i (x^i - x_0^i)$$

Remember $c^i - d^i$ always factorizes as

$$c^i - d^i = \underline{\underline{(c-d)}}\left(c^{i-1} + c^{i-2}d + c^{i-3}d^2 + \cdots + cd^{i-2} + d^{i-1}\right)$$

$$\Longrightarrow G(x) = (x - x_0)\, h(x)$$

where h is a poly with integer coefficients and leading coefficient of $h(x)$ is a_{k+1} and $h(x)$ has degree k. (and remember $p \nmid a_{k+1}$).

Consider the other solutions

$$x_1, \ldots, x_{k+1} \pmod p$$

of $g(x) \equiv 0 \pmod p$. For $j = 1, \ldots, k+1$

$$G(x_j) = g(x_j) - g(x_0) \equiv 0 \pmod p$$

$$= (x_j - x_0)\, h(x_j)$$

$$\Longrightarrow p \mid (x_j - x_0)\, h(x_j)$$

But $p \nmid x_j - x_0$, since $x_0, \ldots, x_{k+1}$ are all distinct modulo p.

$\Rightarrow p \mid h(x_j)$, from Euclid's lemma $p \mid ab \Rightarrow p \mid a$ or $p \mid b$,

$\Rightarrow h(x_j) \equiv 0 \pmod{p}$

for $j = 1, \ldots, k+1$

So summarizing, $h(x)$ has degree k and $k+1$ solutions mod p.

This contradicts our induction assumption above. This came from the assumption in yellow above, namely that the theorem didn't hold for polys of degree $k+1$.

So therefore theorem does hold for polys of degree $k+1$

So by the principle of induction the theorem holds for polys of all degrees $n \geq 1$

So theorem is true.

Continue at 5p.m.

Now, how to solve $f(x) \equiv 0 \mod p_i^{a_i}$

Ex 7.1 Apply theorem 7.2 to obtain all the solutions to

$$f(x) = x^2 + 3x - 16 \equiv 0 \pmod{81}$$

$81 = 3^4$

Bottom level: first solve.

$$f(x) \equiv 0 \pmod 3$$

$$\iff x^2 + 3x - 16 \equiv 0 \pmod 3$$

$$\iff x^2 - 1 \equiv 0 \pmod 3$$

$$\iff x^2 \equiv 1 \pmod 3$$

$$\iff x \equiv 1, 2 \pmod 3$$

First, let $r = 1$, a solution to $f(x) \equiv 0 \pmod 3$

$f'(x) = 2x + 3$, $f'(1) = 5 \not\equiv 0 \pmod 3$

So $r = 1$ will lift uniquely to a solution s for $f(x) \equiv 0 \bmod 9$ given by

$$s = r + tp^{a-1}$$

$$= 1 + t3$$

where t is the unique solution to

$$t \cdot 5 + \frac{-12}{3} \equiv 0 \pmod 3$$

$$\iff 2t - 1 \equiv 0 \pmod 3$$

$$\iff 2t \equiv 1 \pmod 3$$

$$\iff t = 2^{-1} \pmod 3$$

$$= 2$$

So $s = 1 + 2 \cdot 3 = 7$.

So now let $r = 7$, a solution to $f(x) \equiv 0 \mod 9$

$f'(7) = 2 \cdot 7 + 3 = 17 \not\equiv 0 \pmod 9$

So there is a unique associated solution s to $f(x) \equiv 0 \pmod{27}$ given by

$$s = r + t p^{a-1}$$
$$= 7 + t \cdot 9$$

where t is the solution to (7.11)

$$t \cdot 17 + \frac{54}{9} \equiv 0 \pmod 3$$

$$\begin{aligned} f(7) &= 7^2 + 3 \cdot 7 - 16 \\ &= 49 + 21 - 16 \\ &= 54 \end{aligned}$$

$$\iff 2t + 6 \equiv 0 \pmod 3$$
$$\iff 2t \equiv 0 \pmod 3$$
$$\iff t \equiv 0 \pmod 3.$$

So we have the solution $s \equiv 7 \pmod{27}$ to $f(x) \equiv 0 \pmod{27}$.

Now let $r = 7$, a solution $f(x) \equiv 0 \pmod{27}$

$f'(7) = 17 \not\equiv 0 \pmod 3$ so we will have a unique associated solution s to $f(x) \equiv 0 \mod 81$ given by

$$s = r + t p^{a-1}$$
$$= 7 + t \cdot 27$$

where t is the solution to

$$t \cdot 17 + \frac{54}{27} \equiv 0 \pmod 3$$

$$2t + 2 \equiv 0 \pmod 3$$
$$t \equiv -2 \cdot 2^{-1}$$
$$\equiv 2 \pmod 3$$

So we have the solution s.

$$s = 7 + 2 \cdot 27$$
$$= 7 + 54$$
$$= 61 \pmod{81}$$

to the original poly congruence $f(x) \equiv 0 \pmod{81}$

Applying this same lifting procedure to the solution $r = 2$ to $f(x) \equiv 0 \pmod 3$ will produce the second solution

$x \equiv 17$, to $f(x) \equiv 0 \pmod{81}$

seen in Sage output.